

White Paper - OPCP

This white paper gradually describes two main stages of usage of OPCP :

- Ready for use
- Ready for service

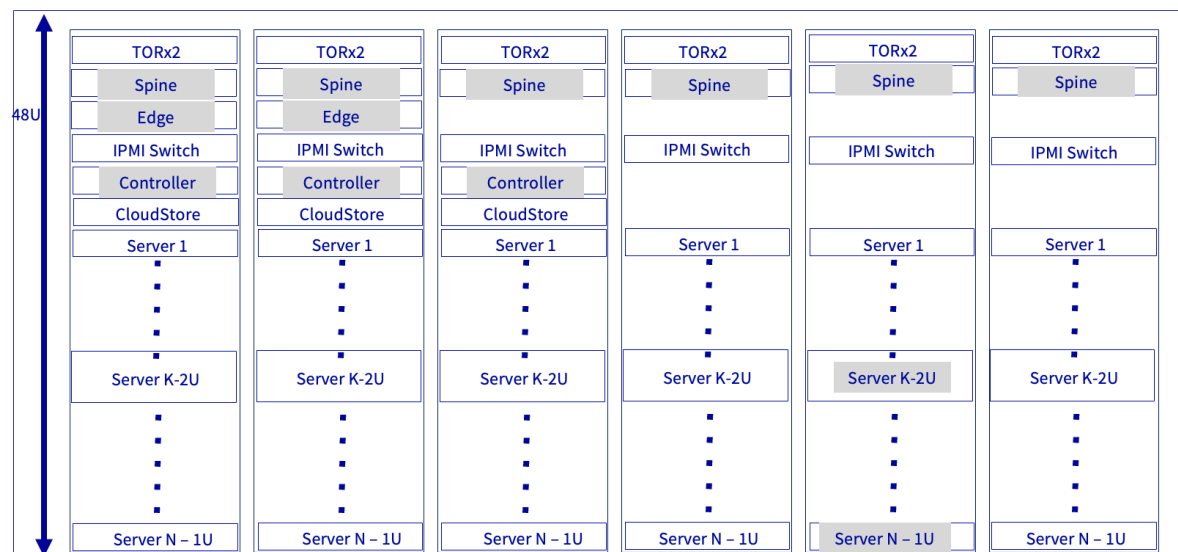
Identification of the Hardware components

At On-Prem-Cloud-Platform reception time, it is needed to identify physically various components. Depending on the configuration chosen, NanoPod, MiniPod, Network Pod or Multiple Network Pod, the racks delivered will present 2 types of configurations :

Network Pod

Network Pod

We distinguish Network components from Management components, from Compute components.



Network components

On Network Pod, there are 3 types of switches : ToR, Spine and Edge

- Two Top of Rack switches (ToR) are present per rack. These ToR have a bandwidth of 25 Gbps per port interface, and their output bandwidth is 6x100 Gbps. Both switches are working like a pair and own their BGP peering.

Note: it is possible to configure port interface bandwidth to 50 Gbps upon request.

- Spines are distributed across racks, with 2 spines at the minimum in total and one Spine at the maximum per rack to ensure rack-fault-tolerance. On the schema above, 6 Spines are present.

- Edges are L2/L3 gateways for OPCP. There are 2 edge switches per Datacenter and therefore per OPCP. Both switches are distributed across racks to ensure rack-fault-tolerance. Edge's role is to connect to the On-premises network and therefore to terminate networks of the OPCP fabric.

Management components

On Network Pod, there are 3 types of management components : Controllers, IPMI Switch and Cloud Store servers.

- Three Controllers are present per Pod. They are distributed across racks to ensure rack fault-tolerance. Controller server is 1U. They are in charge of running OPCP Core Control Plane.
- IPMI Switch is used to connect to each BMC of the compute nodes. IPMI network is not accessible to customer. Instead, data collected at IPMI switch level is used by OPCP Control Plane to trigger actions according to compute node status and IPMI logs.
- Three Cloud Store server is present and can be enabled to deploy Cloud Apps onto compute nodes. Cloud Store servers are redundant across racks. They are in charge of running Cloud Store Control Plane.

Compute components

On Network Pod, servers can be installed throughout the full 48U rack height, excluding the units reserved for Network components and Management components. Minimum server height is 1U. Storage shelves can be added to any server configuration chosen.

Example of a GPU server comprising a storage shelf of 24 disks.

GPU 8 x H200 : 4U

24 x 7.68TB NVME for a raw capacity of 184.32 TB : 2U

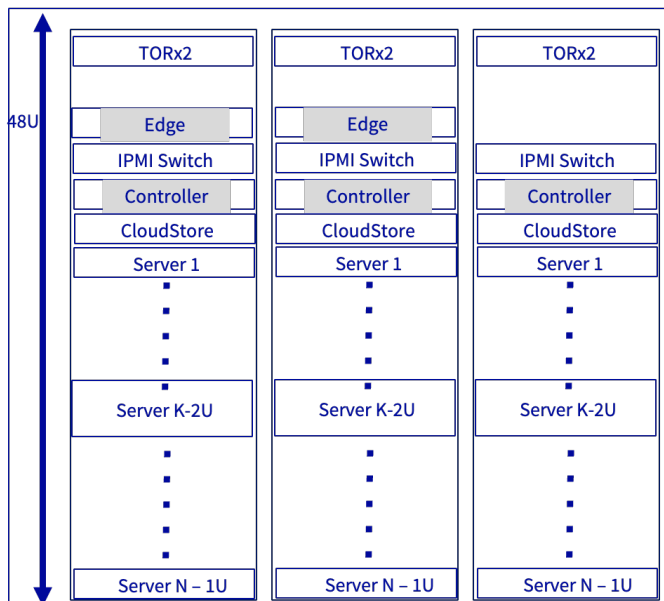
MiniPod

MiniPod is similar to Network Pod.

Except two differences :

- Only 3 racks are present
- No Spine is configured so that TORs connect directly to the Edge switches

The mini pod configuration is not adaptive.

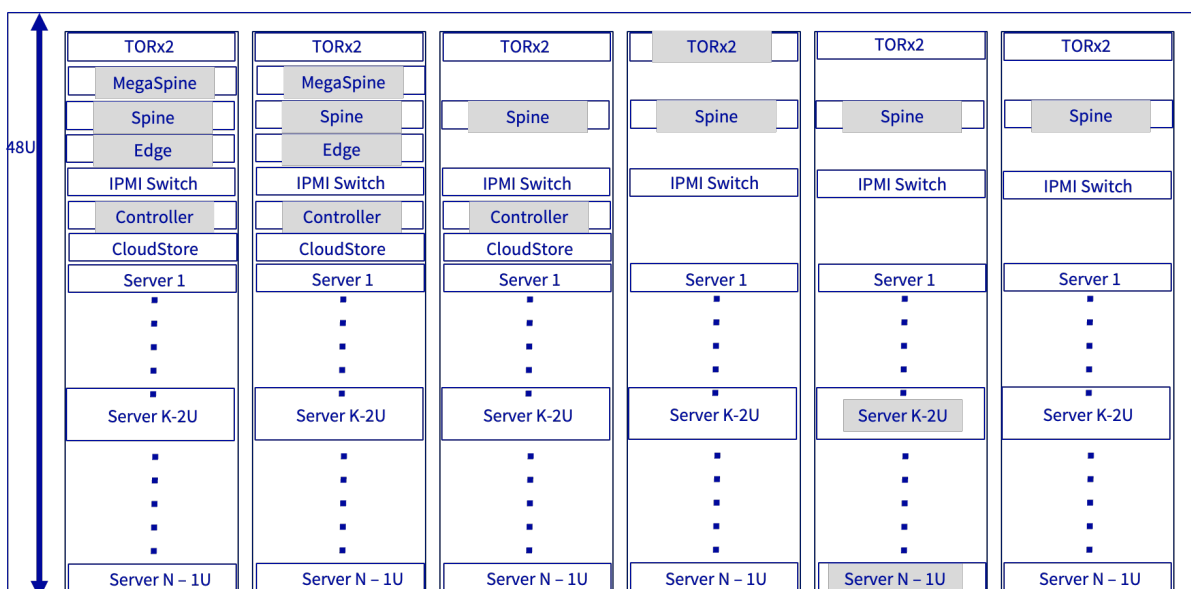


Multiple Network Pod

Multiple Network Pod is similar to Network Pod.

Except two differences :

- Spines are present on each rack
- MegaSpines are present so that Spine connect to MegaSpines within OPCP Fabric

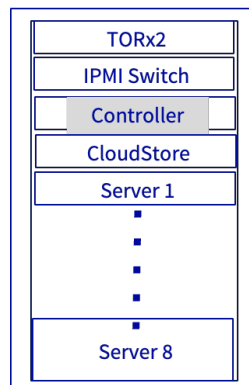


NanoPod

NanoPod is the smallest configuration of OPCP. It comprises 4 to 8 servers and is not adaptive.

Contrary to other OPCP configurations :

- NanoPod comprises only one OPCP Controller and one Cloud Store server
- Edge role is ensured by TORs to connect to the On-Premises network.



Verification of cabling between components

A visual verification of cabling may be needed. It is recommended to start with management component cabling, and then to verify servers cabling.

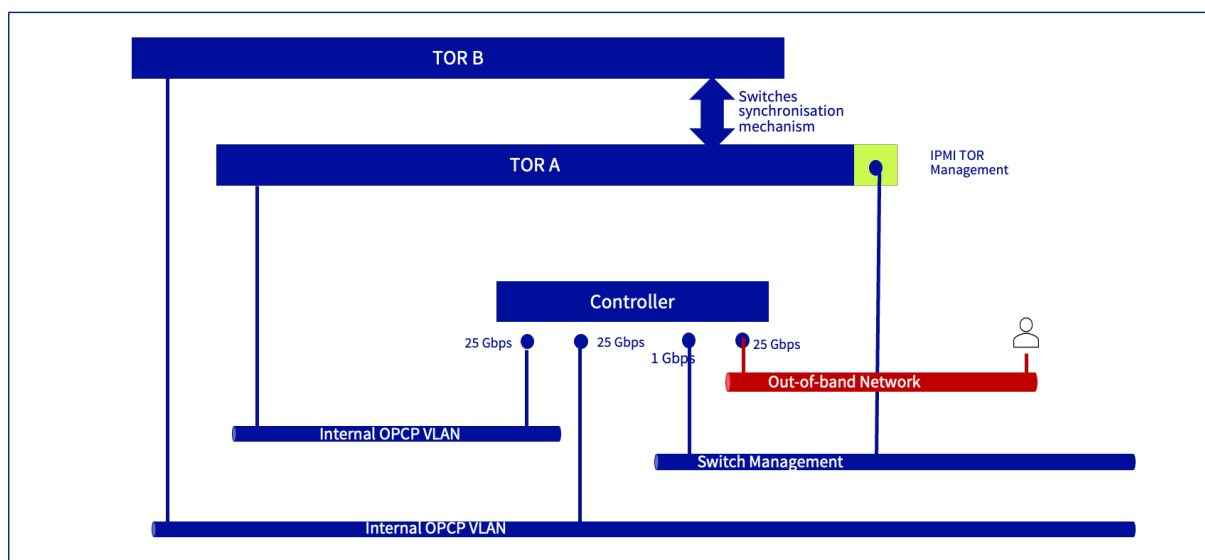
Management component cabling

OPCP Controller is managed through an out-of-band connection on a specific port, represented in red below. When several racks are present, all three controllers are synchronized continuously through OPCP fabric and management of the OPCP Control Plane is to be performed through the out-of-band port interface of each controller.

The controller has two dual-port network cards, hence 4 ports 25 Gbps.

Out of the port targeted to the Out-of-band network, which is left free for connection, there is no possibility to change the destination of the three other ports. These three ports are cabled this way.

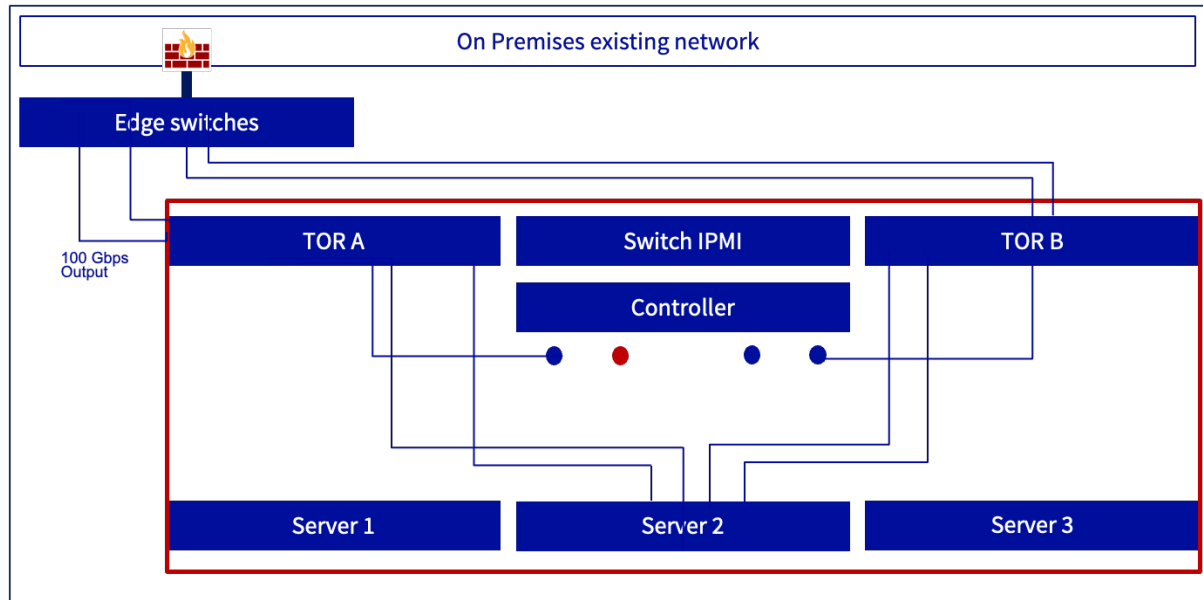
- One port connects to the IPMI ToR management of ToR A.
- Two ports connects to the ToR switches of the rack i.e. ToR A and ToR B:
 - One port interface of first network card connects to ToR A
 - One port interface of second network card connects to ToR B



Servers physical cabling

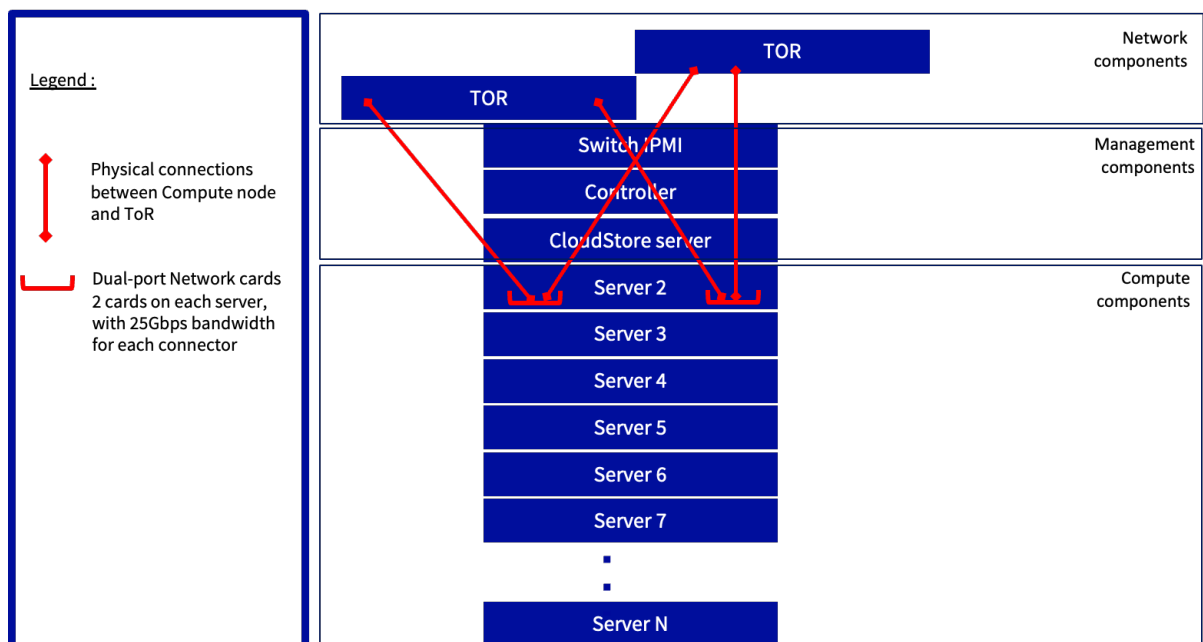
Servers are connected to ToRs through 4 port interfaces.

There is no possibility to change the destination of any of the cables.



Servers have two dual-port network cards, hence 4 ports 25 Gbps.

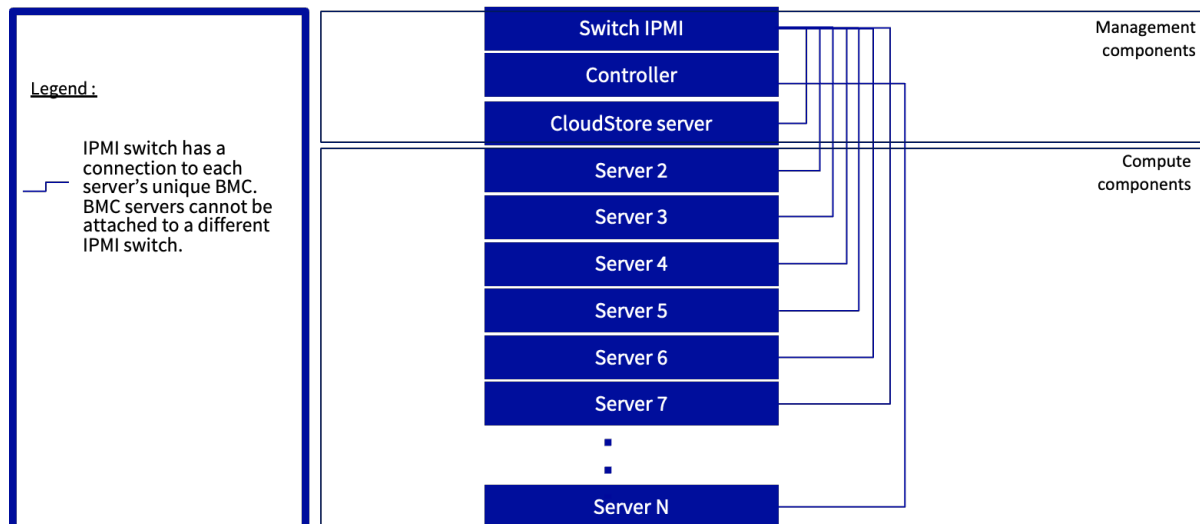
Servers have a 2+2 no-SPOF cross-connections to ToR switches, i.e. each server is connected via 4 physical cables to both TOR switches.



IPMI switch physical cabling

IPMI switch has a connection to each server's unique BMC.

BMC servers cannot be attached to a different IPMI switch, nor IPMI switch can be attached to a tier component which is not racked into OPCP.

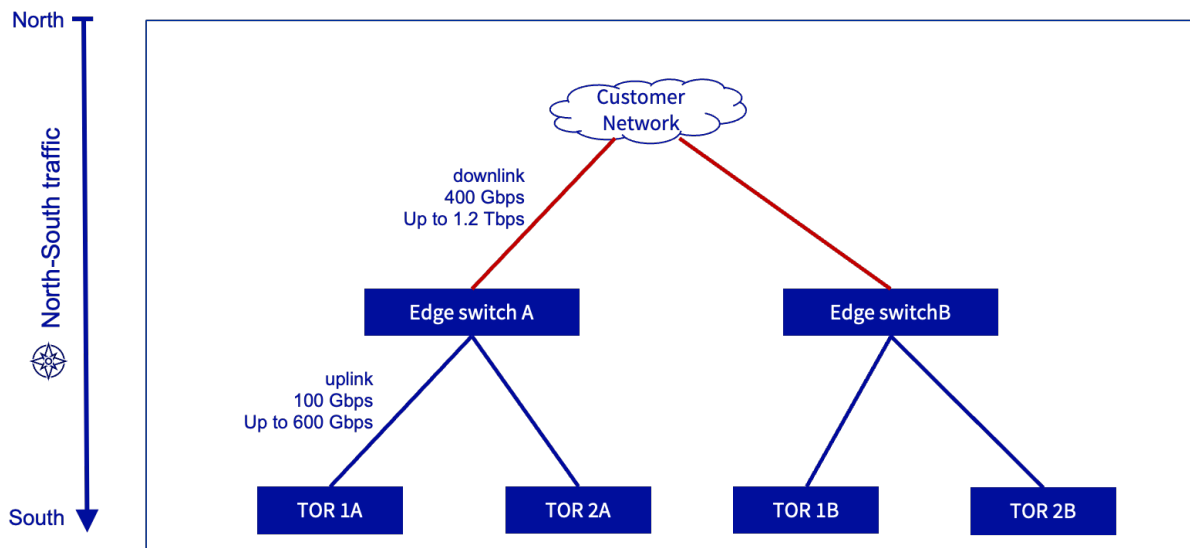


OPCP North-South traffic

Edge cabling and capabilities

North-South traffic i.e. external communication out of OPCP is described here. It is determined by Edge components cabling and bandwidth capabilities.

- Depending on the configuration, either ToR or Spines connect to Edges
- Each ToR connects to only one Edge Switch : ToR A and ToR B on a rack connect respectively to Edge A and Edge B
- ToRs/Spines are connected to Edge switches through an uplink connection of 100Gbps bandwidth at least.
- Depending on the OPCP flavor (Nano, Mini, Network Pod, Multiple Network Pod), several 100 Gbps physical links can be configured to increase uplink bandwidth.



North-South traffic goes through Edge switches.

It is needed when designing the target architecture of OPCP, to determine the bandwidth of the downlink. This allows to evaluate the external traffic which can go out of OPCP, and therefore the hosting capability of OPCP. This is performed thanks to the following elements :

- Edges are connected to each TOR with 100 Gbps links. Uplinks are 6x100Gbps at the maximum. Therefore uplink is an adaptive configuration which depends on expected uplink performance. We define N as the number of physical uplinks from an Edge. N can vary from 2 to 6.
- When needed to connect to external On-premises network, Edge switches have 30 free interfaces of 100Gbps available for this purpose.

Given the number of links from one edge to the number of TORs connected and the number of racks, north-south traffic maximum bandwidth is determined by the formula :

$$\text{Downlink} = N \times R \times 100 \text{ Gbps}$$

where R is the number of racks of OPCP and N is the number of physical uplinks per Edge

Application according to the cabling described in above schema

There are 2 racks and 2 links between an Edge and ToRs :

R = number of racks = 2

N = number of 100 Gbps link from TORs to one Edge = 2

Downlink = $N \times R \times 100 \text{ Gbps} = 400 \text{ Gbps}$

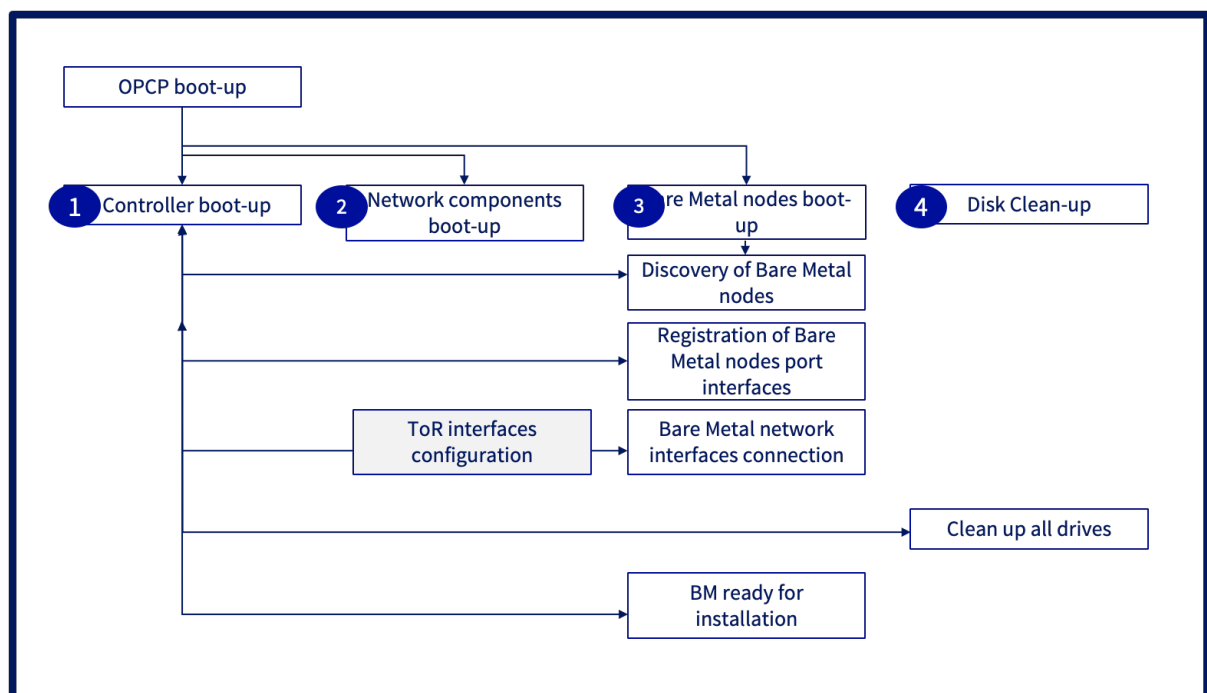
At the maximum, with 6x100 Gbps uplink bandwidth, Downlink can go up to 1.2 Tbps.

200 Gbps ≤ Uplink ≤ 600 Gbps
400 Gbps ≤ Downlink ≤ 1.2 Tbps

OPCP boot-up and autonomous discovery

OPCP will perform all operations below autonomously at boot-up time. Bare Metal nodes are ready for installation at the end of this autonomous phase.

Sequence of boot involves first the management components, then network components, and finally Bare Metal nodes. A disk clean-up is systematically performed at OPCP boot time.



At this point, OPCP is ready for use.

OPCP Configuration

Once boot-up is performed, some minimal configuration steps need to be taken.

Minimal configuration steps

Minimal configuration steps need to be taken to get the best usage of OPCP : CA, OPCP IP admin, DNS, NTP.

Detail of steps to follow :

- Configuration of the Certificate source, internal or external
- Assignment of an IP address to the Control Plane to configure Out-of-band access (external DHCP recommended)
- Registration of OPCP into DNS if resolution on the local domain is required
- Configuration of NTP, with an internal or external source

Once these steps are taken, OPCP endpoints will be accessed by various roles undermentioned.

- admin.<subdomain> : access to the controller UI with highest privileged access
- dashboard.<subdomain> : used by unprivileged users who only access projects

Definition of projects

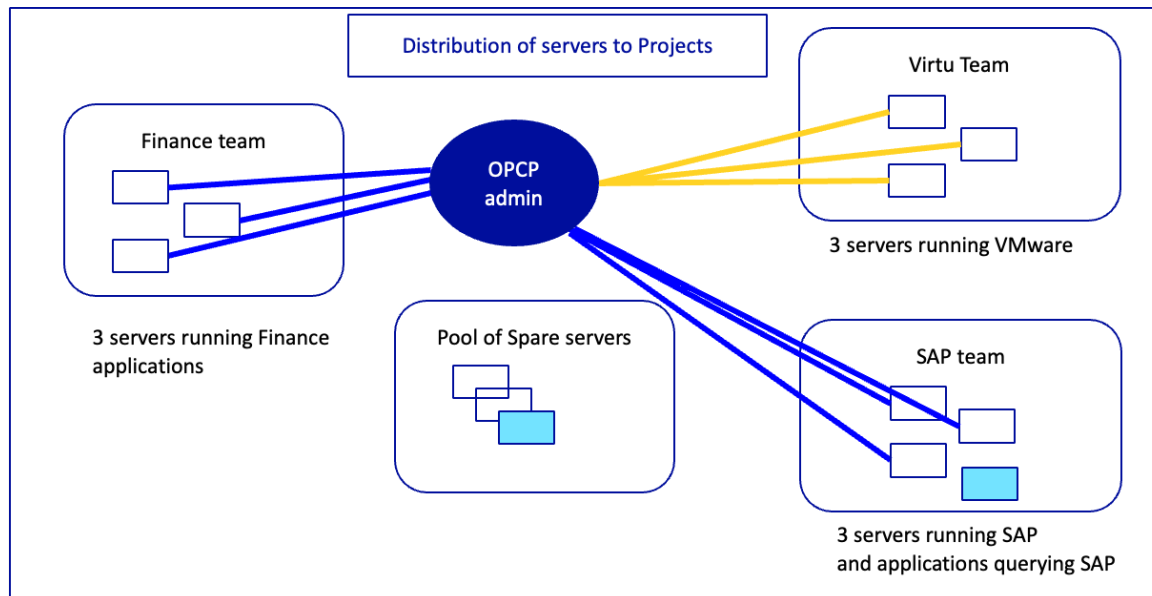
OPCP allows to physically isolate workloads.

In the undermentioned example, OPCP admin assigns servers to different groups across the organization, while ensuring physical isolation of workloads at server level.

A Project identifies a group of servers and is assigned to a different team/organization/end-customer.

OPCP allows to handle several workloads on the same Pod, the breakdown per project is determined by the answer to the question : who is managing this workload ?

As a key element for project distribution, project needs fit with targeted organization.



Creation of roles

OPCP allows to configure a full Identity and Access Management stack, comprising users, groups and roles for OPCP administration and consumption.

As per OPCP best practices, it is recommended to configure distinct roles to segment accesses in production.

DC operator	IT admin	User
-------------	----------	------

What does each role do ?

Manage equipment in the datacenter	Allocate resources to end-users	Consume servers
------------------------------------	---------------------------------	-----------------

Which specific actions are performed by this role ?

- Racks or unracks machines in DC - Fixes servers when a hardware failure occurs - Does the electrical or cooling maintenance - Monitors HW metrics and logs through Grafana and Netbox - Gets its actions tracked and logged onto OPCP Controller	- Creates projects and handles advanced operations on projects - Allocates and deallocate resources for each department / end-user - Creates public custom images and public/private flavors - Enables Cloud Store and deploys cloud infrastructure needed for CloudStore - Creates users and groups on the projects	- Creates networks - Install servers - Create key-pairs - Creates private custom images
--	--	--

Where should it be created ?

Realm Master	Realm Pod	Realm Pod
Who creates it ?		
Super Admin	Super Admin	IT Admin
Which UI access will it get ?		
<a href="https://admin.dashboard.<domain>">https://admin.dashboard.<domain>	<a href="https://dashboard.<domain>">https://dashboard.<domain>	<a href="https://dashboard.<domain>">https://dashboard.<domain>

As a

source of authentication, OPCP allows two possibilities :

- OPCP has built-in support to authenticate users through an existing IdP and established a uni-directional trust to the IdP.
- OPCP allows to declare users locally, and uses reinforced authentication with MFA, OTP, and password policy setup.

OPCP uses realms in order to distinguish domains for OPCP administrators and OPCP users.

In case authentication source is external, a uni-directional trust can be established at realm level, so that

- administrator IDs (IT admin and DC operator) - are authenticated via one IdP
- user IDs (Project users) are authenticated via a different IdP.

Definition of networks

Networks are created at project level, to attach servers which are part of a project to a network. Networks can also be created at administration level to ensure all servers can be configured to attach a port interface to this network.

Protocols used by OPCP are as follows :

- OPCP only uses VLAN to configure networks.
- Neither VXLAN nor Flat VLANs are allowed. It is possible not to set the VLAN ID while creating a network although OPCP will tag the VLAN automatically with a free VLAN ID in this case.
- LACP can be configured on 2 to 4 port interfaces
- Trunk can be used to distinguish several Customer VLANs within one port interface or LACP.

OPCP offers various possibilities allowing to build networks in order to achieve specific connections

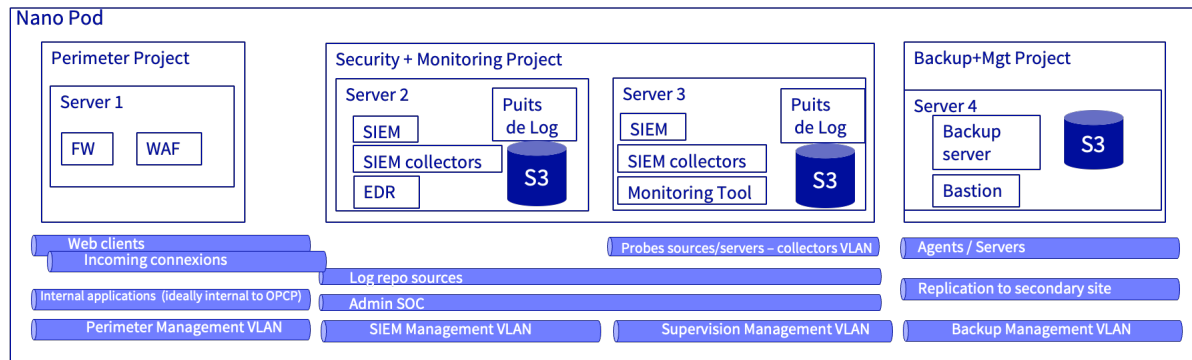
Networks can be shared with all projects

- Networks can be shared with one specific target project
- Networks can be extended on premises, that is to say out of OPCP fabric.

A low-level architecture design will be built. OVHcloud Professional Services can assist you with this and provide you with customized deliverables as LLD and Implementation Plan.

Example of a Mega-Bastion hosted on NanoPod

For instance, as a use case built on OPCP, we may consider cybersecurity tools needing to be hosted with physical isolation between critical workloads.



OPCP allows to host these workloads in a highly secured manner.

In this example, we face three main specifications described below.

- 1/ Bastion is needed to manage other cybersecurity applications, then Management VLAN cannot be configured on the same project and same Baremetal nodes than Web clients and Incoming connexions
- 2/ The SIEM, which is the SOC tool, cannot become unavailable. SIEM collectors and Log Repository need to be available in case of server failure to avoid log collection to be stopped or logs to be dropped. Therefore SIEM is defined with High Availability on 2 servers within a project
- 3 / Given low data criticality, Bastion can though fit onto the Backup Project.

Creation of instances

OPCP provides a list of validated OS. Any other distribution and Operating system can be added to the Image repository by providing a QCOW2 image, built for an Openstack installation.

The list of validated OS is :

- Debian 11
- Debian 12 (LACP available at boot)
- CentOS 9
- Ubuntu 24 (LACP available at boot)

Creation of an OS instance is the same process as a installing a virtual machine on an Overcloud Openstack :

- Select a flavor
- Select an image
- Select network(s) to connect the instance to
- Select an SSH-KEY pair to be able to connect remotely to the instance. In case no SSH-KEY is provided, console available on the OPCP UI will be used.

RAID can be configured with any level of RAID compliant with the number of available drives.

Encryption is automatically managed during this step.

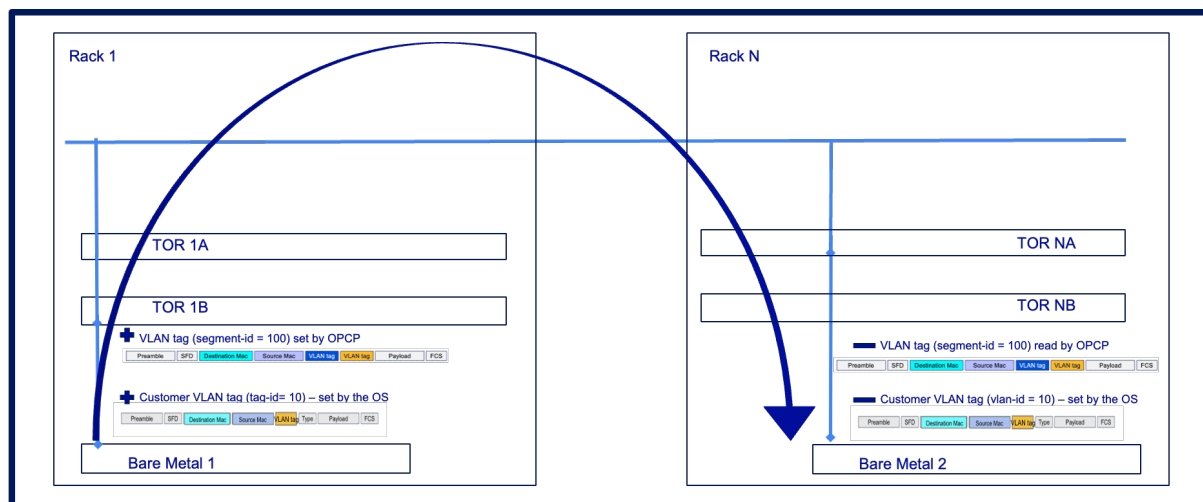
At this point, OPCP is ready for service.

OPCP Network segmentation in cross-rack communication use case

OPCP allows to scale-out network segmentation thanks to built-in network features.

When an ethernet frame moves from one Bare Metal server to another, in any OPCP rack, OPCP allows to tag traffic at Operating System level with a dedicated VLAN ID identifying the specific OS network interface. Q-in-Q and Trunk are used so that inner VLANs use the same outer VLAN, defined by OPCP.

In addition to this, OPCP uses an underneath mechanism to extend the number of possible VLANs transiting through OPCP fabric to several millions.

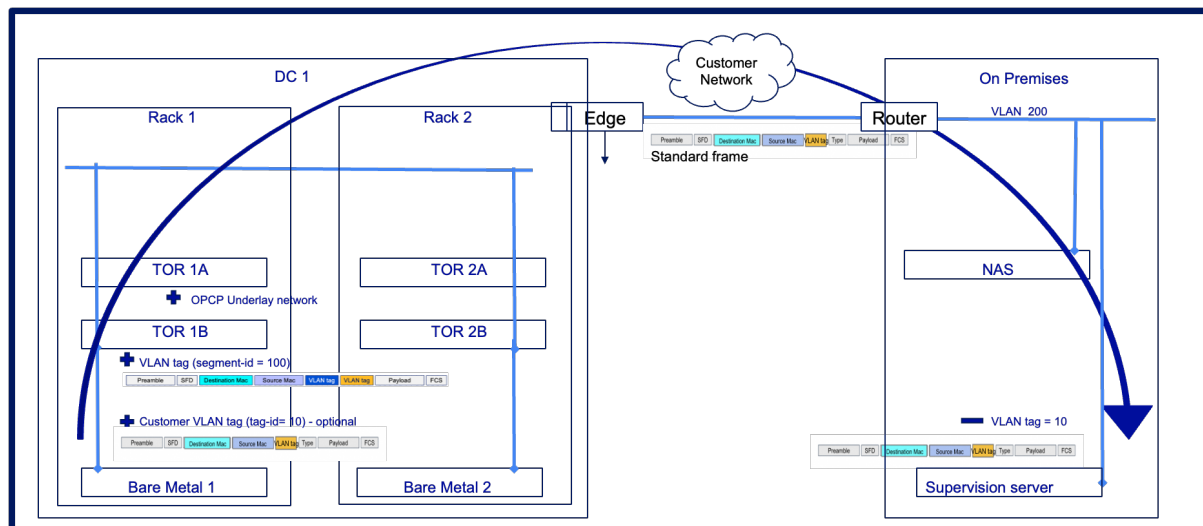


These network segmentation functionalities are built-in and are not configurable.

Edge role in communication with On Premises use case

When on-premises equipments need to be reached, OPCP allows to address tier equipments through the Edges, upon configuration.

In the example below, Edge switches act as a L2 Gateway to reach out to the On-premises supervision Center on VLAN 10.



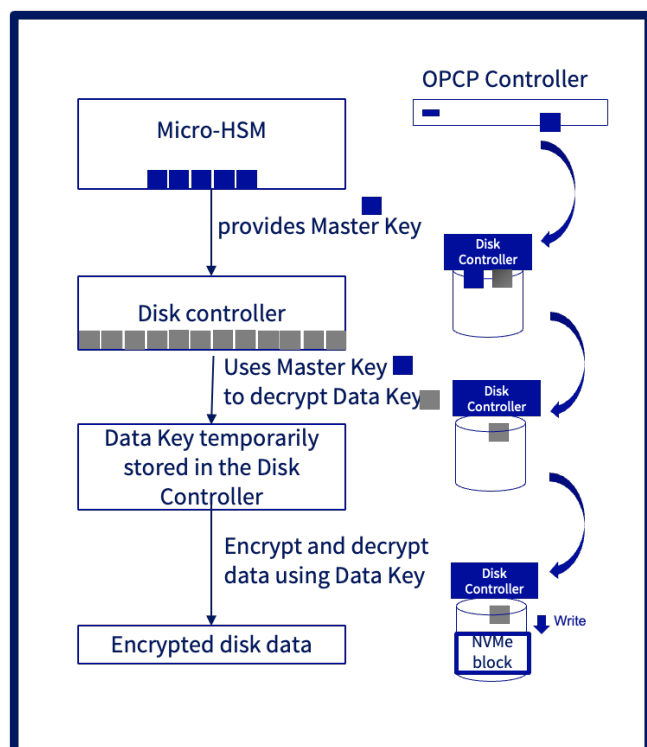
Data-at-rest encryption

OPCP has built-in support to handle data-at-rest encryption.

Data encryption on OPCP is unbreakable and cannot be deactivated

Main characteristics of data-at-rest encryption performed by OPCP are as follows :

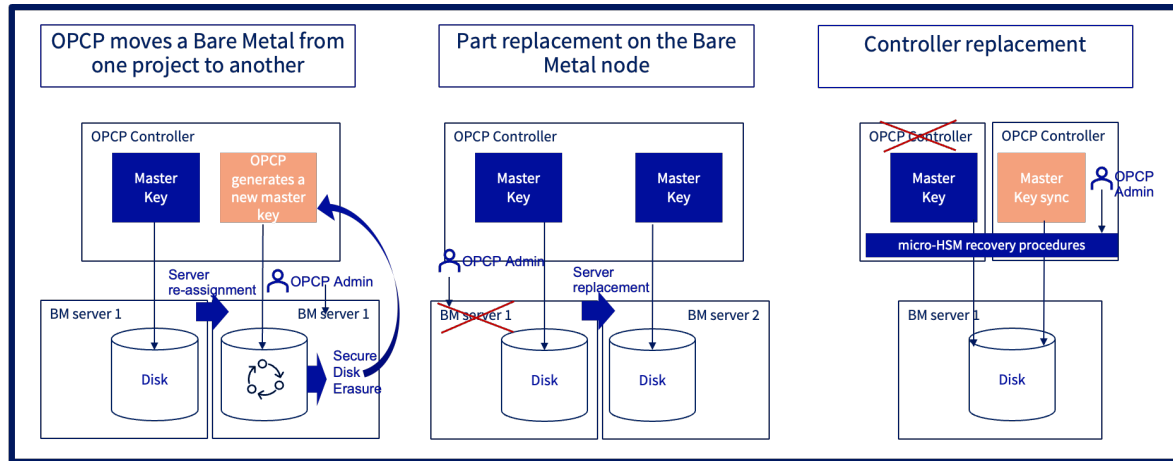
- Disk cannot be moved and read maliciously out of OPCP. A disk taken out of OPCP and inserted in another system cannot be read.
- Encryption takes place at Disk Controller level. The attack surface of CPU/RAM vulnerabilities is reduced because the CPU does not handle encryption tasks.
- Disks encrypts data during write operations.



These data-at-rest functionalities are built-in and are not configurable.

Data protection during maintenance operations

OPCP allows maintenance operations to be performed seamlessly while preserving data protection objective.



When a Bare Metal node is moved from one project to another, OSCP performs a complete clean-up of all Bare Metal node drives, to prevent data from being read maliciously.

On purpose, Data is preserved in case of Bare Metal node component failure, and controller failure events.